WILLIAM ALEXAKIS

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EDUCATION

St. Catherine's British School — **Athens, Greece**
International Baccalaureate Diploma Programme — **September 2026 — June 2028**
Computer Science HL, Mathematics Analysis & Approaches HL, Physics HL

IGCSE Programme — **September 2024 — June 2026**
Computer Science, Mathematics Extended, Physics Triple Award

PROJECTS

self-attention.c — **June 2026**
github.com/williamalexakis/self-attention.c
- Wrote a scaled dot-product self-attention implementation in C with zero dependencies in ~100 LOC, featuring Q/K dot product computation, score normalization by root of d_k to prevent softmax saturation, softmax weight distribution, and weighted value aggregation.

Phase Programming Language — **August 2025 — December 2025**
github.com/williamalexakis/phase
- Wrote a statically-typed programming language in C with zero dependencies in ~5,000 LOC, featuring a 25-opcode stack-based virtual machine, 5 primitive types, and tuple destructuring.
- Built a full compilation pipeline including a lexer, recursive-descent parser, static type checker, bytecode generator, and a stack-based virtual machine with scoped call frames.
- Implemented 23 error types with source-mapped diagnostics displaying exact file/line/column, visual markers, and diff-style fix suggestions.

ESSAYS

williamalexakis.github.io/essays/mathematics-of-scaled-dot-product-self-attention/ — **June 2026**
williamalexakis.github.io/essays/chunk-based-file-reading-in-c/ — **June 2026**
williamalexakis.github.io/essays/bytecode-interpretation-from-scratch-in-c/ — **December 2025**

EXTRACURRICULAR ACTIVITIES

Open-Source Software Contributions — **June 2026 — Present**
Contributor
- Implemented syntax error handler with source references for 6 error types in C++ for a programming language, reducing redundant code and improving error UX (github.com/omnimistic/algo-lang/pull/13).

StCatsHacks Line Robot Competition — **December 2023 — December 2025**
Team Leader & Lead Programmer
- Won 1st place for 3 consecutive years (2023 — 2025) leading a 3-person team.
- Designed a 3-sensor line-following algorithm to resolve simultaneous sensor activation on acute-angle turns, outpacing the standard PID approach used by competing teams on both speed and accuracy.

ADDITIONAL INFORMATION

Technical Skills
C, Python, Git

Spoken Languages
Fluent English, Native Greek